A two-dimensional angular-locus classification for Property P

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Source Question

Mandal–Manna–Paul–Sain define $x \perp_B^\varepsilon y$ by

$$\|x + \lambda y\|^2 \geq \|x\|^2 - 2\varepsilon \|x\| \, \|\lambda y\| \qquad (\lambda \in \mathbb{R})$$

and say that a bounded linear operator T preserves ε-orthogonality at every point if

$$x \perp_B y \quad \Longrightarrow \quad Tx \perp_B^\varepsilon Ty \qquad (x, y \in X).$$

Their Property P asks whether, for a given Banach space X, some $\varepsilon \in (0,1)$ forces every such $T \in \mathcal{L}(X)$ to be a scalar multiple of an isometry [1]. The source question and definition appear here:

In view of the Blanco-Koldobsky-Turnšek characterization of isometries, it is natural to consider the following more general query:

Given a Banach space $\mathbb{X}$, does there exist an $\epsilon \in (0,1)$, such that any $T \in \mathbb{L}(\mathbb{X})$ preserving ϵ-orthogonality is an isometry multiplied by a constant?

It is clear that an affirmative answer to the above query will directly lead to a proper refinement of the Blanco-Koldobsky-Turnšek theorem, which is the main motivation behind this study. In order to address the above query, it is convenient to introduce the following definition:

Definition 1.2. Let $\mathbb{X}$ be a Banach space. We say that $\mathbb{X}$ has the *Property P* if there exists an $\epsilon \in (0,1)$ such that the following two conditions are equivalent:
(i) $T \in \mathbb{L}(\mathbb{X})$ preserves ϵ-orthogonality at each $x \in \mathbb{X}$.
(ii) $T \in \mathbb{L}(\mathbb{X})$ is a scalar multiple of an isometry.

Clearly, whenever a Banach space $\mathbb{X}$ possesses the *Property P*, a refinement of the Blanco-Koldobsky-Turnšek theorem is possible for $\mathbb{X}$. On the other hand, if $\mathbb{X}$ does not possess the *Property P*, it follows that the Blanco-Koldobsky-Turnšek theorem can not be improved by considering approximate preservation of orthogonality. Our main aim is to produce concrete examples of both types of Banach spaces by considering certain geometric properties of the concerned space. We first investigate some geometric and analytic properties of finite-dimensional polyhedral Banach spaces, including the smoothness and the number of extreme supporting functionals of each points, related to such preservation. Additionally, we analyze the linear operators satisfying such preservation of orthogonality, in connection with the geometry of the concerned spaces. As an application of this study, we characterize the isometries on some Banach spaces, thus obtaining refinements of the Blanco-Koldobsky-Turnšek theorem on these spaces. Finally, we establish that a two-dimensional Banach space, whose unit sphere is given by a regular $2n$-gon ($n > 2$), and the sequence space ℓ_∞^n ($n > 2$) do satisfy the *Property P*. On the other hand, we prove that the sequence spaces ℓ_p ($1 \le p < \infty$) do not possess this property.

We end this section with the following characterization of the approximate Birkhoff-James orthogonality, which will be used repeatedly.

Theorem 1.3. [4, Th. 2.4] *Let $\mathbb{X}$ be a Banach space and let $x, y \in \mathbb{X}$. Let $\epsilon \in [0,1)$. Then x is ϵ-approximate Birkhoff-James orthogonal to y if and only if there exists $f \in J(x)$ such that $|f(y)| \le \epsilon \|y\|$.*

2. Main Results

We begin this section with the following proposition.

Proposition 2.1. *Let $\mathbb{X}$ be a finite-dimensional polyhedral Banach space. For each $f \in Ext\ B_{\mathbb{X}^*}$, consider the set $\mathcal{A}_f = \{x \in \mathbb{X} : J(x) = \{f\}\}$. Then the following results hold true:*

This packet gives a claimed complete answer for real two-dimensional Banach spaces.

Definitions and Tools

For $0 \neq x \in X$, put

$$J(x) = \{f \in X^* : \|f\| = 1,\ f(x) = \|x\|\}.$$

The projective angular locus is

$$\mathcal{A}_X = \{[x] \in \mathbb{P}(X) : J(x) \text{ is not a singleton}\}.$$

Thus $[x] \in \mathcal{A}_X$ exactly when the norm is nonsmooth in the direction x.

We use the source's support-functional characterization:

$$u \perp_B^{\varepsilon} v \quad \iff \quad \exists f \in J(u) \text{ such that } |f(v)| \le \varepsilon \|v\|$$

[1, Theorem 1.3]. We also use the exact Blanco–Koldobsky–Turnsek theorem: a linear operator preserving exact Birkhoff–James orthogonality is a scalar multiple of an isometry [3, 4].

Finally, we use Chmielinski–Stypka's bounded-below theorem for the same approximate preservation relation. If an additive map $T : X \to Y$ between real normed spaces satisfies

$$x \perp_B y \Longrightarrow Tx \perp_B^{\varepsilon} Ty,$$

then T is linear and bounded and

$$(1 - 8\varepsilon)\|T\|\,\|x\| \le \|Tx\| \le \|T\|\,\|x\|, \qquad x \in X$$

[2, Theorem 4.4]. We apply only the displayed estimate, and only to linear operators.

Proof Intuition

The classification is governed by projective first-order rigidity. Suppose Property P fails. Then there are norm-one nonsimilarities T_n preserving ε_n-orthogonality with $\varepsilon_n \downarrow 0$. The Chmielinski–Stypka estimate makes a subsequential limit an isometry; composing it away gives $S_n \to I$. Dividing S_n by its distance to the scalar line gives a non-scalar first-order operator A.

At a nonsmooth direction a, the support set $J(a)$ is a nontrivial segment. If A moves the projective line $[a]$ to first order, then $[S_n a]$ approaches $[a]$ from one side. Planar convexity says that all support functionals at $S_n a$ then collapse to the corresponding one-sided endpoint of $J(a)$. Choosing an exact orthogonality direction from the opposite endpoint gives a contradiction to approximate preservation. Hence every angular line must be an eigenline of A. Three projective eigenlines force A to be scalar, contradicting the normalization.

The converse is also geometric. If there are at most two angular lines, a small diagonal perturbation can fix all of them while stretching one transverse direction. Such perturbations are not similarities, but because they fix the only possible support discontinuities, they preserve exact orthogonality up to an arbitrarily small Chmielinski error.

Lemma 1 (One-sided support collapse in a normed plane)**.** *Let X be a real two-dimensional Banach space, let $a \in S_X$, and suppose $J(a)$ is not a singleton. Choose one of the two local sides of the projective line $[a]$. There is an endpoint j_+ of the segment $J(a)$ such that whenever $x_n \in S_X$ tends projectively to a from that side and $f_n \in J(x_n)$, every norm-convergent subsequence of f_n converges to j_+. The other side gives the other endpoint j_-.*

Proof. This is the standard one-dimensional subdifferential fact for a planar convex body. We include the reduction for reference. In an affine chart around a, the boundary of B_X on either side of a can be represented as the graph of a concave function ϕ of one real variable. Supporting functionals at a boundary point correspond exactly to slopes in the subdifferential interval of ϕ at the corresponding parameter. For a concave function on an interval, the left and right derivative functions are monotone, and the subdifferential intervals $\partial\phi(t_n)$ converge to the appropriate one-sided derivative as $t_n \to 0^+$ or $t_n \to 0^-$. Under the affine identification this one-sided derivative is precisely one endpoint of $J(a)$. This proves the claim. ☐

Lemma 2 (Infinitesimal preservers fix angular lines)**.** *Let X be a real two-dimensional Banach space. Suppose $\varepsilon_n \downarrow 0$, $S_n \to I$, and each S_n preserves ε_n-orthogonality at every point. Assume that*

$$S_n = \lambda_n I + d_n A_n, \qquad \lambda_n \to 1, \quad d_n \downarrow 0, \quad A_n \to A.$$

If $[a] \in \mathcal{A}_X$, then $Aa \in \mathbb{R}a$.

Proof. Suppose $Aa \notin \mathbb{R}a$ for some unit representative a. After passing to a subsequence, the projective points $[S_n a]$ approach $[a]$ from one fixed side. Let j_+ be the corresponding endpoint of $J(a)$ from Lemma 1. Choose another support functional $j_0 \in J(a)$, $j_0 \neq j_+$, and choose $0 \neq y \in \ker j_0$. Then $a \perp_B y$, while $j_+(y) \neq 0$, because two distinct support functionals at a have distinct kernels in a plane.

Since S_n preserves ε_n-orthogonality, there is $g_n \in J(S_n a)$ such that

$$|g_n(S_n y)| \leq \varepsilon_n \|S_n y\|.$$

Passing to a subsequence, g_n converges in S_{X^*}. By Lemma 1, the only possible limit is j_+. Letting $n \to \infty$ in the displayed inequality gives $j_+(y) = 0$, a contradiction. Therefore $Aa \in \mathbb{R}a$. □

Lemma 3 (Three angular lines imply Property P)**.** *If a real two-dimensional Banach space X has at least three projective angular lines, then X has Property P.*

Proof. Assume Property P fails. Then for some $\varepsilon_n \downarrow 0$ there are norm-one operators T_n preserving ε_n-orthogonality at every point which are not scalar multiples of isometries. By the Chmielinski–Stypka estimate,

$$(1 - 8\varepsilon_n)\|x\| \leq \|T_n x\| \leq \|x\|.$$

After passing to a subsequence, $T_n \to U$, and the displayed estimate shows that U is a linear isometry. Replacing T_n by $S_n = U^{-1}T_n$, we may assume $S_n \to I$. Isometric conjugation preserves the same approximate orthogonality condition, and S_n is still not a scalar multiple of an isometry.

Let

$$d_n = \operatorname{dist}(S_n, \mathbb{R}I) > 0$$

and choose $\lambda_n \in \mathbb{R}$ with $\|S_n - \lambda_n I\| = d_n$. Since $S_n \to I$, we have $\lambda_n \to 1$ and $d_n \to 0$. Put

$$A_n = d_n^{-1}(S_n - \lambda_n I).$$

After a subsequence, $A_n \to A$. Moreover $\operatorname{dist}(A, \mathbb{R}I) = 1$, so A is not scalar.

Lemma 2 says that every angular projective line is an eigenline of A. But a non-scalar linear operator on a two-dimensional real vector space has at most two projective eigenlines. This contradicts the existence of three angular lines. Hence Property P holds. □

Lemma 4 (Finite angular stabilizer perturbations)**.** *Let X be a real two-dimensional Banach space with finite angular locus $\mathcal{A}_X$. Let $R_n \to I$ be invertible operators such that every line in $\mathcal{A}_X$ is invariant under every R_n. Then for every $\varepsilon > 0$, all sufficiently large n have the property that R_n preserves ε-orthogonality at every point.*

Proof. It is enough to prove that the optimal Chmielinski error tends to zero. If not, there are $\eta > 0$, unit vectors x_n, y_n, and a subsequence, still called R_n, such that $x_n \perp_B y_n$ but

$$\min_{g \in J(R_n x_n)} \frac{|g(R_n y_n)|}{\|R_n y_n\|} \geq \eta.$$

Pass to a further subsequence with $x_n \to x$ and $y_n \to y$. The exact orthogonality relation is closed, so $x \perp_B y$.

For each n, choose $f_n \in J(x_n)$ with $f_n(y_n) = 0$. We will find $g_n \in J(R_n x_n)$ with $g_n(R_n y_n) \to 0$, contradicting the displayed lower bound.

If x is smooth, then the support-functional map is continuous in a neighborhood of x avoiding the finite set $\mathcal{A}_X$, so the unique functionals in $J(R_n x_n)$ converge to the same limit as f_n. Hence $g_n(R_n y_n) \to 0$.

Now suppose $[x] \in \mathcal{A}_X$. Because $\mathcal{A}_X$ is finite, a punctured projective neighborhood of $[x]$ contains no other angular line. After passing to a subsequence, either x_n lies on $[x]$ for all n, or $[x_n]$ approaches $[x]$ from one fixed side. In the first case, $[R_n x_n] = [x]$, and for large n the scalar on this invariant line is positive; taking $g_n = f_n$ gives $g_n \in J(R_n x_n)$ and

$$g_n(R_n y_n) = g_n(y_n) + g_n((R_n - I)y_n) \to 0.$$

In the second case, R_n fixes $[x]$ and is close to the identity, so $[R_n x_n]$ approaches $[x]$ from the same side. By the one-sided support collapse lemma, both f_n and the unique member $g_n \in J(R_n x_n)$ tend to the same endpoint of $J(x)$. Therefore

$$|g_n(R_n y_n)| \le |(g_n - f_n)(R_n y_n)| + |f_n((R_n - I)y_n)| + |f_n(y_n)| \to 0.$$

This final contradiction proves the lemma. □

Lemma 5 (At most two angular lines imply failure of Property P)**.** *If a real two-dimensional Banach space X has at most two projective angular lines, then X does not have Property P.*

Proof. Let L_1, L_2 be two independent projective lines containing $\mathcal{A}_X$; if $\mathcal{A}_X$ has fewer than two elements, choose the missing line or lines arbitrarily. Pick unit vectors $e_i \in L_i$. For $t > 0$, define

$$R_t e_1 = e_1, \qquad R_t e_2 = (1+t)e_2.$$

Then $R_t \to I$, and R_t leaves every angular line invariant. By Lemma 4, for every $0 < \varepsilon < 1$, all sufficiently small $t > 0$ make R_t preserve ε-orthogonality at every point.

However R_t is not a scalar multiple of an isometry for $t \neq 0$. Indeed, if $R_t = cU$ with U an isometry, then from the unit vector e_1 we get $|c| = 1$, while from e_2 we get $|c| = 1 + t$, impossible. Thus no ε can witness Property P. □

Theorem 1 (Two-dimensional classification)**.** *Let X be a real two-dimensional Banach space. Then X has Property P if and only if its projective angular locus $\mathcal{A}_X$ contains at least three projective lines.*

Proof. The positive implication is Lemma 3. The negative implication is Lemma 5. □

Consequences

For a two-dimensional polyhedral space, the angular lines are exactly the projective vertex lines. Hence the theorem recovers the previous polyhedral classification: a polygonal unit ball has Property P exactly when it has at least six extreme points. The theorem also handles non-polyhedral examples with infinitely many corners; any such plane has Property P as soon as it has three distinct angular directions.

Smooth two-dimensional spaces fail Property P because $\mathcal{A}_X = \varnothing$. This agrees with the separate lane-16 uniformly smooth packet.

Verification and Novelty Check

Local indexes were checked for arXiv:2501.10719, Property P, two-dimensional polyhedral classifications, finite angular locus, and Birkhoff–James orthogonality. Web searches on July 19, 2026 located the source arXiv record, the published Colloquium Mathematicum record, the 2026 Lebesgue–Bochner continuation, and Chmielinski–Stypka's 2025 bounded-below theorem. No exact two-dimensional angular-locus classification was found.

Human-review recommendation: focus first on Lemma 1 and its use in Lemmas 2 and 4. Those are the only genuinely delicate planar convexity steps. The remaining argument is compactness, first-order projective rigidity, and elementary diagonal perturbations.

References

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